IoT C Programs for Online Compiler

These are **simulation programs in standard C**, so they will run in online compilers such as Programiz, OnlineGDB, JDoodle, and similar tools.

Use the figures below as the exact diagrams to draw in your paper. The code is intentionally very short.

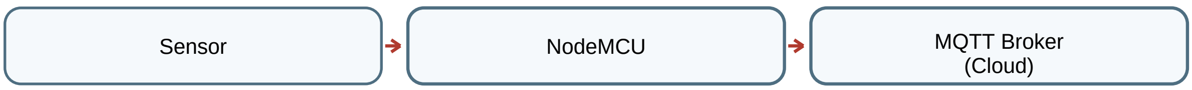
Lab Note: In real hardware, NodeMCU uses ESP8266 libraries. For online compilers, these C programs simulate the same logic.

1. MQTT Publisher

Online-compiler C simulation of publishing sensor data to an MQTT topic.

Figure to draw: **Sensor** → **NodeMCU** → **MQTT Broker (Cloud)**

Sketch



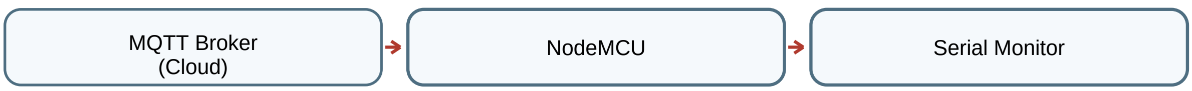

```
#include <stdio.h>
int main() {
    int sensor;
    printf("Enter sensor value: ");
    scanf("%d", &sensor);
    printf("Connecting to MQTT broker...\n");
    printf("Publishing to topic: iot/data\n");
    printf("Message sent: %d\n", sensor);
    return 0;
}
```

2. MQTT Subscriber

Online-compiler C simulation of subscribing to a topic and receiving a message.

Figure to draw: **MQTT Broker (Cloud)** → **NodeMCU** → **Serial Monitor**

Sketch



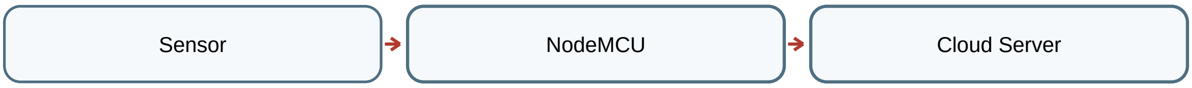

```
#include <stdio.h>
int main() {
    char msg[50];
    printf("Subscribing to topic: iot/data\n");
    printf("Enter received message: ");
    scanf("%s", msg);
    printf("Message received: %s\n", msg);
    return 0;
}
```

3. HTTP Client using GET and POST

Online-compiler C simulation of sending sensor data by HTTP GET and POST.

Figure to draw: **Sensor** → **NodeMCU** → **Cloud Server**

Sketch



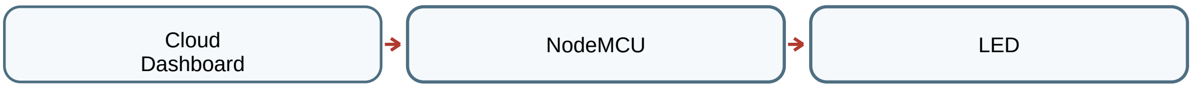

```c
#include <stdio.h>
int main() {
    int value;
    printf("Enter sensor value: ");
    scanf("%d", &value);
    printf("GET: http://server.com/get?value=%d\n", value);
    printf("POST Data: value=%d\n", value);
    return 0;
}
```

4. Cloud-Triggered Actions

Online-compiler C simulation of cloud command controlling an output device.

Figure to draw: **Cloud Dashboard** → **NodeMCU** → **LED**

Sketch



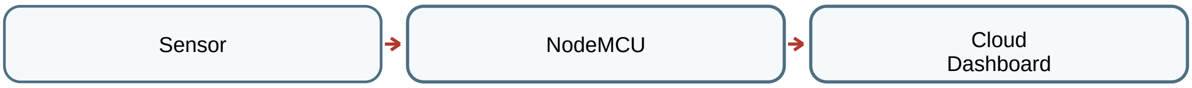

```c
#include <stdio.h>
int main() {
    int command;
    printf("Enter command (1=ON, 0=OFF): ");
    scanf("%d", &command);
    if (command == 1) printf("LED turned ON\n");
    else printf("LED turned OFF\n");
    return 0;
}
```

5. Real-time Data Streaming

Online-compiler C simulation of continuous sensor values sent to a dashboard.

Figure to draw: **Sensor** → **NodeMCU** → **Cloud Dashboard**

Sketch




```c
#include <stdio.h>
int main() {
    int data[5], i;
    printf("Enter 5 sensor values:\n");
    for(i=0; i<5; i++) scanf("%d", &data[i]);
    printf("Streaming data:\n");
    for(i=0; i<5; i++) printf("%d ", data[i]);
    return 0;
}
```